

- 1 On a test, Zac's mark corresponded to a z-score of 2. The test scores had a mean of 63 and a standard deviation of 8.

What was Zac's actual mark on the test?

- A 65 B 67
C 73 D 79

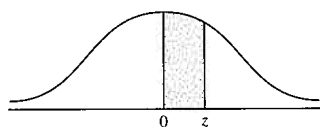
(1 mark)

(Q8, 2021 HSC) **Medium**

- 2 A random variable is normally distributed with mean 0 and standard deviation 1. The table gives the probability that this random variable lies between 0 and z for different values of z .

z	0.1	0.2	0.3	0.4	0.5	0.6
Probability	0.0398	0.0793	0.1179	0.1554	0.1915	0.2257

The probability values given in the table for different values of z are represented by the shaded area in the following diagram.



- a Using the table, show that the probability that a value from a random variable that is normally distributed with mean 0 and standard deviation 1 is greater than 0.3 is equal to 0.3821. (1 mark) **Hard**
- b Birth weights are normally distributed with a mean of 3300 grams and a standard deviation of 570 grams. By first calculating a z-score, find how many babies, out of 1000 born, are expected to have a birth weight greater than 3471 grams.

(3 marks)

CQ (Q38, 2021 HSC) **Hard**

- 3 In a particular city, the heights of adult females and the heights of adult males are each normally distributed.

Information relating to two females from that city is given in Table 1.

Table 1

Height	Gender	Percentage of females in this city shorter than this person
175 cm	Female	97.5%
160.6 cm	Female	16%

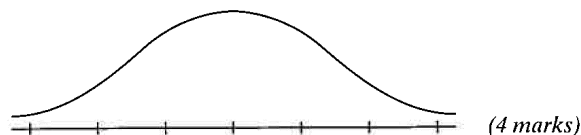
The means and standard deviations of adult females and males, in centimetres, are given in Table 2.

Table 2

	Mean	Standard deviation
Females	μ	σ
Males	1.05μ	1.1σ

A selected male is taller than 84% of the population of adult males in this city.

By first labelling the normal distribution curve below with the heights of the two females given in Table 1, calculate the height of the selected male, in centimetres, correct to two decimal places.



CQ (Q41, 2021 HSC) **Hard**

- 4 John recently did a class test in each of three subjects. The class scores on each test were normally distributed.

The table shows the subjects and John's scores as well as the mean and standard deviation of the class scores on each test.

Subject	John's score	Mean	Standard deviation
French	82	70	8
Commerce	80	65	5
Music	74	50	12

Relative to the rest of the class, which row of the table below shows John's strongest subject and his weakest subject?

	Strongest subject	Weakest subject
A	Commerce	French
B	French	Music
C	Music	French
D	Commerce	Music

(1 mark)

CQ (Q8, 2020 HSC) **Medium**

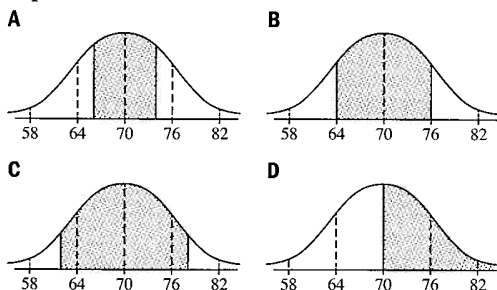
- 5 The Intelligence Quotient (IQ) scores for adults in City A are normally distributed with a mean of 108 and a standard deviation of 10. The IQ scores for adults in City B are normally distributed with a mean of 112 and a standard deviation of 16.

- a Yin is an adult who lives in City A and has an IQ score of 128. What percentage of the adults in this city have an IQ score higher than Yin's? (2 marks) **Medium**
- b There are 1000000 adults living in City B. Calculate the number of adults in City B that would be expected to have an IQ score lower than Yin's. (2 marks) **Medium**

- c Simon, an adult who lives in City A, moves to City B. The z -score corresponding to his IQ score in City A is the same as the z -score corresponding to his IQ score in City B. By first forming an equation, calculate Simon's IQ score. Give your answer correct to one decimal place. (3 marks) **Hard**

(Q35, 2020 HSC)

- 6 The scores on an examination are normally distributed with a mean of 70 and a standard deviation of 6. Michael received a score on the examination between the lower quartile and the upper quartile of the scores. Which shaded region most accurately represents where Michael's score lies?



(1 mark)

(Q15, 2019 HSC) **Hard**

- 7 In a particular country, the birth weight of babies is normally distributed with a mean of 3000 grams. It is known that 95% of these babies have a birth weight between 1600 grams and 4400 grams.

One of these babies has a birth weight of 3497 grams. What is the z -score of this baby's birth weight? (2 marks)

(Q38, 2019 HSC) **Medium**

- 8 A set of data is normally distributed with a mean of 48 and a standard deviation of 3. Approximately what percentage of the scores lies between 39 and 45?

- A 15.85% B 31.7%
C 47.5% D 49.85% (1 mark)

(Q23, 2018 HSC) **Medium**

- 9 Joanna sits a Physics test and a Biology test.
i Joanna's mark in the Physics test is 70. The mean mark for this test is 58 and the standard deviation is 8. Calculate the z -score for Joanna's mark in this test. (1 mark) **Easy**
ii In the Biology test, the mean mark is 64 and the standard deviation is 10. Joanna's z -score is the same in both the Physics test

and the Biology test. What is her mark in the Biology test? (2 marks) **Easy**

(Q27e, 2018 HSC)

- 10 The heights of Year 12 girls are normally distributed with a mean of 165 cm and a standard deviation of 5.5 cm.

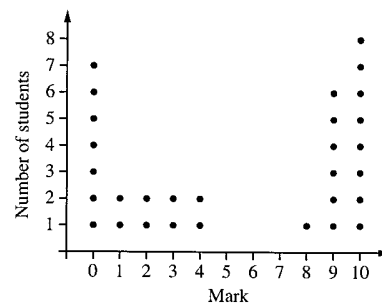
What is the z -score for a height of 154 cm?

- A -2 B -0.5
C 0.5 D 2 (1 mark)

(Q13, 2017 HSC) **Easy**

- 11 All the students in a class of 30 did a test.

The marks, out of 10, are shown in the dot plot.



- i Find the median test mark. (1 mark) **Medium**
ii The mean test mark is 5.4. The standard deviation of the test marks is 4.22. Using the dot plot, calculate the percentage of the marks which lie within one standard deviation of the mean. (2 marks) **Hard**
iii A student states that for any data set, 68% of the scores should lie within one standard deviation of the mean. With reference to the dot plot, explain why the student's statement is NOT relevant in this context. (1 mark) **Hard**

(Q29d, 2017 HSC)

- 12 The speed limit outside a school is 40 km/h. Year 11 students measured the speed of passing vehicles over a period of time. They found the set of data to be normally distributed with a mean speed of 36 km/h and a standard deviation of 2 km/h.

What percentage of the vehicles passed the school at a speed greater than 40 km/h?

- A 2.5% B 5%
C 47.5% D 95% (1 mark)

(Q13, 2016 HSC) **Easy**

- 13 A machine produces cylindrical pipes. The mean of the diameters of the pipes is 8 cm and the standard deviation is 0.04 cm.

Assuming a normal distribution, what percentage of cylindrical pipes produced will have a diameter less than 796 cm?

- A 16% B 32%
C 34% D 68% (1 mark)

(Q20, 2015 HSC) Medium

- 14 The results of two tests are normally distributed. The mean and standard deviation for each test are displayed in the table.

	Mathematics	English
$\bar{x}$	70	75
s	6.5	8

Kristoff scored 74 in Mathematics and 80 in English. He claims that he has performed better in English.

Is Kristoff correct? Justify your answer using appropriate calculations. (2 marks)

(Q28b, 2015 HSC) Medium

- 15 The weights of 10000 newborn babies in NSW are normally distributed. These weights have a mean of 3.1 kg and a standard deviation of 0.35 kg.

How many of these newborn babies have a weight between 2.75 kg and 4.15 kg?

- A 4985 B 6570
C 8370 D 8385 (1 mark)

(Q24, 2014 HSC) Medium

- 16 There are 60000 students sitting a state-wide examination. If the results form a normal distribution, how many students would be expected to score a result between 1 and 2 standard deviations above the mean?

You may assume for normally distributed data that:

- 68% of scores have z -scores between minus -1 and 1
- 95% of scores have z -scores between minus -2 and 2
- 99.7% of scores have z -scores between minus -3 and 3 .

- A 8100 B 16200
C 20400 D 28500 (1 mark)

(Q20, 2013 HSC) Hard

- 17 A machine produces nails. When the machine is set correctly, the lengths of the nails are normally distributed with a mean of 6.000 cm and a standard deviation of 0.040 cm.

To confirm the setting of the machine, three nails are randomly selected. In one sample the lengths are 5.950, 5.983 and 6.140.

The setting of the machine needs to be checked when the lengths of two or more nails in a sample lie more than 1 standard deviation from the mean.

Does the setting on the machine need to be checked? Justify your answer with suitable calculations. (2 marks)

(Q29b, 2012 HSC) Medium

- 18 Two brands of light bulbs are being compared. For each brand, the life of the light bulbs is normally distributed.

Life of light bulbs (in hours)

	Mean	Standard deviation
Brand A	450	25
Brand B	500	50

- i One of the Brand B light bulbs has a life of 400 hours.

What is the z -score of the life of this light bulb? (1 mark) Easy

- ii A light bulb is considered defective if it lasts less than 400 hours. The following claim is made:

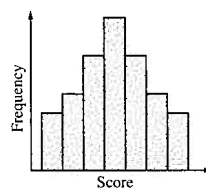
'Brand A light bulbs are more likely to be defective than Brand B light bulbs.'

Is this claim correct? Justify your answer, with reference to z -scores or standard deviations or the normal distribution. (2 marks) Medium

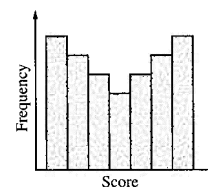
(Q27c, 2011 HSC)

- 19 Which of the following frequency histograms shows data that could be normally distributed?

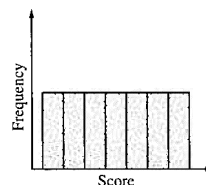
A



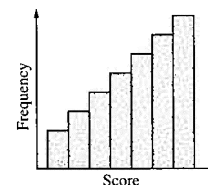
B



C



D



(1 mark)

(Q4, 2010 HSC) Easy

- 20 The marks in a class test are normally distributed. The mean is 100 and the standard deviation is 10.

- i Jason's mark is 115. What is his z -score? (1 mark) Easy

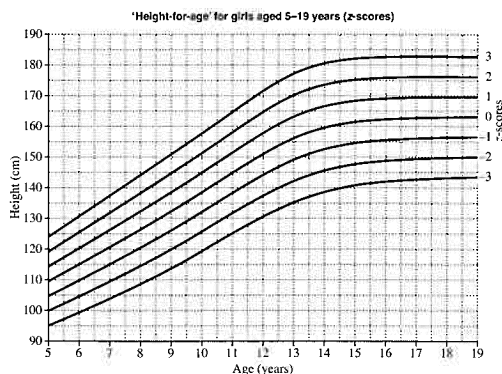
- ii Mary has a z-score of 0. What mark did she achieve in the test? (1 mark) **Easy**
- iii What percentage of marks lie between 80 and 110?
You may assume the following:
- 68% of marks have z-scores between -1 and 1
 - 95% of marks have z-scores between -2 and 2
 - 99.7% of marks have z-scores between -3 and 3. (2 marks) **Medium**

(Q24c, 2010 HSC)

- 21 In Broken Hill, the maximum temperature for each day has been recorded. The mean of these maximum temperatures during spring is 25.8°C , and their standard deviation is 4.2°C .
- i What temperature has a z-score of -1? (1 mark) **Easy**
- ii What percentage of spring days in Broken Hill would have maximum temperatures between 21.6°C and 38.4°C ?
You may assume that these maximum temperatures are normally distributed and that
- 68% of maximum temperatures have z-scores between -1 and 1
 - 95% of maximum temperatures have z-scores between -2 and 2
 - 99.7% of maximum temperatures have z-scores between -3 and 3. (3 marks) **Medium**

(Q25d, 2009 HSC)

- 22 The following graph indicates z-scores of 'height-for-age' for girls aged 5–19 years.



- i What is the z-score for a six year old girl of height 120 cm? (1 mark) **Easy**
- ii Rachel is $10\frac{1}{2}$ years of age.
- If 2.5% of girls of the same age are taller than Rachel, how tall is she? (1 mark) **Medium**
 - Rachel does not grow any taller. At age $15\frac{1}{2}$, what percentage of girls of the same age will be taller than Rachel? (2 marks) **Medium**

- iii What is the average height of an 18 year old girl? (1 mark) **Medium**
- iv For adults (18 years and older), the Body Mass Index is given by $b = \frac{m}{h^2}$ where
 m = mass in kilograms and h = height in metres.
The medically accepted healthy range for B is $21 \leq B \leq 25$.
What is the minimum weight for an 18 year old girl of average height to be considered healthy? (2 marks) **Medium**

(Q28a, 2008 HSC)

- 23 The results of two class tests are normally distributed. The means and standard deviations of the tests are displayed in the table.

	Test 1	Test 2
Mean	60	58
Standard deviation	6.2	6.0

- i Stuart scored 63 in Test 1 and 62 in Test 2. He thinks that he has performed better in Test 1. Do you agree? Justify your answer using appropriate calculations. (2 marks) **Medium**
- ii If 150 students sat for Test 2, how many students would you expect to have scored less than 64? (2 marks) **Hard**

(Q25d, 2007 HSC)

- 24 A clubhouse uses four long-life light globes for five hours every night of the year. The purchase price of each light globe is \$6.00 and they each cost \$ d per hour to run.

The manufacturer's specifications state that the expected life of the light globes is normally distributed with a standard deviation of 170 hours. What is the mean life, in hours, of these light globes if 97.5% will last up to 5000 hours? (1 mark)

(Q27b iv, 2007 HSC) **Medium**

- 25 In a normally distributed set of scores, the mean is 23 and the standard deviation is 5.

Approximately what percentage of the scores will lie between 18 and 33?

- A 34% B 47.5%
C 68% D 81.5% (1 mark)

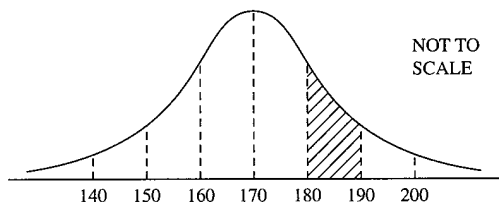
(Q17, 2006 HSC) **Medium**

- 26 The weights of boxes of Brekky Bicks are normally distributed. The mean is 754 grams and the standard deviation is 2 grams.

- i What is the z-score of a box of Brekky Bicks with a weight of 754 g? (1 mark) **Easy**
- ii What is the weight of a box that has a z-score of -1 ? (1 mark) **Medium**
- iii Brekky Bicks boxes are labelled as having a weight of 750 g. What percentage of boxes will have a weight less than 750 g? (2 marks) **Medium**

(Q26c, 2005 HSC)

- 27 The normal distribution shown has a mean of 170 and a standard deviation of 10.



- i Roberto has a raw score in the shaded region. What could his z-score be? (1 mark) **Medium**
- ii What percentage of the data lies in the shaded region? (2 marks) **Medium**

(Q24c, 2004 HSC)

- 28 Results for an aptitude test are given as z-scores. In this test, Hardev gains a z-score of 1.
 - i Interpret Hardev's score with reference to the mean and standard deviation of the test. (2 marks) **Medium**
 - zii The scores for the test are normally distributed. What proportion of people sitting the test obtain a higher score than Hardev? (1 mark) **Medium**

(Q25c, 2003 HSC)

- 29 Results for an aptitude test are given as z-scores. In this test Di gained a z-score of 3. The test has a mean of 55 and a standard deviation of 6.

What was Di's actual mark in this test?

- A 57 B 58
C 64 D 73 (1 mark)

(Q8, 2002 HSC) **Medium**

- 30 In the town of Burrow the ages of the residents are normally distributed. The mean age is 40 years and the standard deviation is 12 years.

Approximately what percentage of the residents are younger than 52?

- A 16% B 32%
C 68% D 84% (1 mark)

(Q12, 2002 HSC) **Medium**

- 31 A factory produces bags of flour. The weights of the bags are normally distributed, with a mean of 900 g and a standard deviation of 50 g.

What is the best approximation for the percentage of bags that weigh more than 1000 g?

- A 0% B 2.5%
C 5% D 16% (1 mark)

(Q19, 2001 HSC) **Medium**

- 32 Results for a reading test are given as z-scores. In this test, Kim gained a z-score equal to -2 .

- i Interpret this z-score in terms of the mean and standard deviation of the test. (2 marks) **Easy**
- ii If the test has a mean of 75 and a standard deviation of 5, calculate the actual mark scored by Kim. (1 mark) **Medium**

(Q23b, 2001 HSC)

- 1 Zac's mark was two standard deviations above the mean.

$$\text{Zac's mark} = 63 + 2 \times 8 = 79$$

Answer D

- 2 a From the table,

$$P(0 < z < 0.3) = 0.1179$$

$$\text{Now } P(z > 0) = 0.5$$

$$\text{So } P(z > 0.3) = 0.5 - 0.1179 = 0.3821$$

(1 mark)

b $\mu = 3300, \sigma = 570,$
 $x = 3471$

$$z = \frac{x - \mu}{\sigma} = \frac{3471 - 3300}{570} = 0.3$$

$$P(\text{birth weight} > 3471 \text{ g})$$

$$= 0.3821 \text{ (from a)}$$

$$\text{Expected number in 1000 births}$$

$$= 0.3821 \times 1000$$

$$= 382.1$$

So, out of 1000 babies born, 382 would be expected to have a birth weight greater than 3471 g. (3 marks)

- 3 68% of scores have z-scores between -1 and 1.

34% have z-scores between -1 and 0.

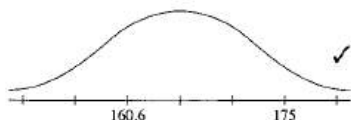
$(50 - 34)\% = 16\%$ will have z-scores less than -1.

So a height of 160.6 cm for a female corresponds to a z-score of -1.

95% of scores have z-scores between -2 and 2.

2.5% have z-scores less than -2, meaning 97.5% have z-scores less than 2.

So a height of 175 cm for a female corresponds to a z-score of 2.



There are three standard deviations between $z = -1$ and $z = 2$.

$$3\sigma = 175 - 160.6$$

$$= 14.4$$

$$\sigma = 4.8$$

$$\mu = 160.6 + 4.8 = 165.4$$

$$\text{Now, } 1.05\mu = 1.05 \times 165.4 = 173.67$$

$$1.1\sigma = 1.1 \times 4.8 = 5.28$$

So the mean for males is 173.67 and the standard deviation is 5.28.

84% of scores are less than a z-score of 1.

So the height of a male who is taller than 84% of the population corresponds to a z-score of 1.

$$\text{Height} = (173.67 + 5.28) \text{ cm} = 178.95 \text{ cm}$$

(4 marks)

4 French: $z = \frac{82 - 70}{8} = 1.5$

Commerce: $z = \frac{80 - 65}{5} = 3$

Music: $z = \frac{74 - 50}{12} = 2$

So Commerce is strongest and French weakest.

Answer A

5 a City A: $x = 128, \mu = 108, \sigma = 10$

$$z = \frac{x - \mu}{\sigma} = \frac{128 - 108}{10} = 2$$

Now 95% of scores are within two standard deviations of the mean.

So 5% are more than two standard deviations from the mean.

Half of these are above two standard deviations.

So 2.5% of adults will have an IQ higher than Yin's. (2 marks)

b City B: $x = 128, \mu = 112, \sigma = 16$

$$z = \frac{x - \mu}{\sigma} = \frac{128 - 112}{16} = 1$$

Now 68% of scores are within one standard deviation of the mean.

Half of these, 34%, are between the mean and one standard deviation above the mean.

50% are below the mean.

So 84% of adults will have an IQ lower than Yin's.

Number of adults = 84% of 1 000 000

840 000 adults would be expected to have an IQ lower than Yin's. (2 marks)

- c Let x be Simon's IQ.

$$\text{Now } \frac{x - 108}{10} = \frac{x - 112}{16}$$

$$16(x - 108) = 10(x - 112)$$

$$16x - 1728 = 10x - 1120$$

$$6x = 608$$

$$x = 101.3333333...$$

$$= 101.3 \text{ (1 d.p.)}$$

So Simon's IQ would be 101.3 to one decimal place. (3 marks)

6 $\mu = 70, \sigma = 6$

So scores between 64 and 76 are within 1 standard deviation of the mean.

Now 68% of scores are within 1 standard deviation of the mean.

50% lie between the upper and lower quartiles.

So the shaded region can only be that in option A

Answer A

7 $\mu = 3000 \text{ g}$

In normally distributed data 95% are within two standard deviations of the mean.

But 95% of these babies lie between 1600 g and 4400 g.

So two standard deviations is 1400 g.

$$\therefore \sigma = 700 \text{ g}$$

$$z = \frac{x - \mu}{\sigma} = \frac{3497 - 3000}{700} = 0.71$$

(2 marks)

8 $\mu = 48, \sigma = 3$

So 45 is one and 39 is three standard deviations below the mean.

Now 99.7% of scores lie within three standard deviations of the mean.

68% of scores lie within one standard deviation of the mean.

$\therefore 99.7\% - 68\%$ or 31.7% lie between one and three standard deviations.

Half of these are below the mean.

Required percentage

$$= 0.5 \times 31.7\%$$

$$= 15.85\%$$

Answer A

- 9 i Physics: $x = 70$, $\mu = 58$,
 $\sigma = 8$

$$z = \frac{x - \mu}{\sigma} = \frac{70 - 58}{8} = 1.5$$

(1 mark)

- ii Biology: $z = 1.5$, $\mu = 64$,
 $\sigma = 10$

$$z = \frac{x - \mu}{\sigma} \Rightarrow 1.5 = \frac{x - 64}{10}$$

$$15 = x - 64$$

$$x = 79$$

Joanna's mark in Biology is 79.

(2 marks)

- 10 $x = 154$, $\mu = 165$, $\sigma = 5.5$

$$z = \frac{x - \mu}{\sigma} = \frac{154 - 165}{5.5} = -2$$

Answer A

- 11 i median = $\frac{4+8}{2} = 6$

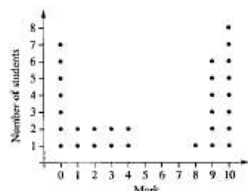
(1 mark)

- ii $\mu = 5.4$, $\sigma = 4.22$

$$\mu + \sigma = 5.4 + 4.22 = 9.62$$

$$\mu - \sigma = 5.4 - 4.22 = 1.18$$

So scores within one standard deviation of the mean are scores greater than 1.18 but less than 9.62.



From the dot plot there are 13 such scores.

$$\begin{aligned} \text{\% within 1 standard deviation of the mean} &= \frac{13}{30} \times 100\% \\ &= 43\frac{1}{3}\% \end{aligned}$$

(2 marks)

- iii 68% of scores lie within one standard deviation of the mean if the data is normally

distributed. Normally distributed data has a bell-shaped graph. From the dot plot it is obvious that the distribution is not bell-shaped and the data is not normally distributed.

(1 mark)

- 12 40 km/h = 36 km/h + 2 × 2 km/h
So it is 2 standard deviations above the mean.

95% lie within 2 standard deviations of the mean so 5% lie outside that. Half of those are above the mean.

So 2.5% of the vehicles will have a speed greater than 40 km/h.

Answer A

- 13 8 - 0.04 = 7.96

So a diameter of 7.96 cm is one standard deviation below the mean.

Now 68% of scores are within one standard deviation of the mean.

Percentage more than one standard deviation from the mean = (100 - 68)% or 32%

Half of those will be more than one standard deviation below the mean.

So the percentage with a diameter less than 7.96 cm is (32 ÷ 2)% or 16%.

Answer A

- 14 Mathematics: $z = \frac{x - \mu}{\sigma} = \frac{74 - 70}{6.5} = 0.6153846\ldots$

$$\begin{aligned} \text{English: } z &= \frac{x - \mu}{\sigma} = \frac{80 - 75}{8} = 0.625 \end{aligned}$$

So, yes Kristoff is correct. The z-score is slightly higher for English than for Mathematics so he performed slightly better in English.

(2 marks)

- 15 3.1 - 2.75 = 0.35

So a weight of 2.75 kg is 1 standard deviation below the mean.

Half of 68%, or 34%, of babies will have weights between 2.75 kg and 3.1 kg.

$$4.15 - 3.1 = 1.05$$

$$1.05 \div 0.35 = 3$$

So a weight of 4.15 kg is

3 standard deviations above the mean.

Half of 99.7%, or 49.85%, of babies will have weights between 3.1 kg and 4.15 kg.

$$\text{Now } 34\% + 49.85\% = 83.85\%$$

$$83.85\% \text{ of } 10000 = 8385$$

Answer D

- 16 68% of scores have z-scores between -1 and 1.

Half of these, 34%, will have z-scores between 0 and 1.

95% have z-scores between -2 and 2.

So 47.5% will have z-scores between 0 and 2.

47.5% - 34% or 13.5% have z-scores between 1 and 2.

$$13.5\% \text{ of } 60000 = 8100$$

So 8100 students would be expected to score a result between 1 and 2 standard deviations above the mean.

Answer A

- 17 $\mu = 6.000$ $\sigma = 0.040$

$$\mu + \sigma = 6.000 + 0.040 = 6.040$$

$$\mu - \sigma = 6.000 - 0.040 = 5.960$$

So the lengths of two or more of the 3 nails should be between 5.960 cm and 6.040 cm.

$$\text{Now } 5.950 < 5.960$$

$$\text{and } 6.140 > 6.040$$

So the lengths of two nails are more than one standard deviation from the mean.

Yes, the machine does need to be checked.

(2 marks)

- 18 i $x = 400$, $\mu = 500$, $\sigma = 50$

$$z = \frac{x - \mu}{\sigma} = \frac{400 - 500}{50} = -2$$

(1 mark)

- ii Brand A: $x = 400$, $\mu = 450$, $\sigma = 25$

$$z = \frac{x - \mu}{\sigma} = \frac{400 - 450}{25} = -2$$

The z-scores for light bulbs with a life of 400 hours is the same for both brands. This means the same percentage of light bulbs are likely to be defective. The claim is not correct.

(2 marks)

- 19 Normally distributed data has a bell-shaped curve.

Answer A

20 $\mu = 100, \sigma = 10$

i $x = 115$

$$z = \frac{x - \mu}{\sigma} = \frac{115 - 100}{10} = 1.5$$

Jason's z-score is 1.5.

(1 mark)

- ii A z-score of 0 represents the mean. So Mary scored 100 in the test.

(1 mark)

- iii 90 is 1 standard deviation below the mean and 110 is 1 standard deviation above the mean.

So 68% of marks are between 90 and 110.

80 is 2 standard deviations below the mean.

Percentage of marks with z-scores between 1 and 2 or between -1 and -2

$$= 95\% - 68\%$$

$$= 27\%$$

Half of these have z-scores between -1 and -2.

Percentage of marks between 80 and 90

$$= 27\% \div 2$$

$$= 13.5\%$$

Total percentage of marks between 80 and 110

$$= 13.5\% + 68\%$$

$$= 81.5\%$$

✓ (2 marks)

21 i $z = \frac{x - \mu}{\sigma}$

$$\therefore -1 = \frac{x - 25.8}{4.2}$$

$$-4.2 = x - 25.8$$

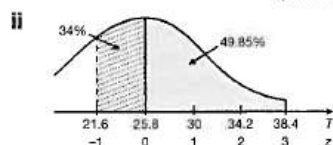
$$x = -4.2 + 25.8$$

$$= 21.6^\circ\text{C}$$

OR

Temperature with a z-score of -1 is one standard deviation below the mean, i.e. $25.8 - 4.2 = 21.6^\circ\text{C}$.

(1 mark)



Shaded area from 25.8°C to 38.4°C is for z-scores between 0 and 3.

This is $99.7\% \div 2 = 49.85\%$ of scores.

Dotted area from 21.6°C to 25.8°C is for z-scores between 0 and -1.

This is $68\% \div 2 = 34\%$ of scores.

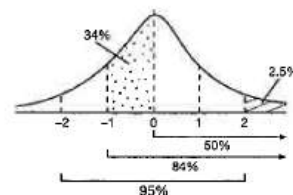
$\therefore$ Percentage of days with maximum temperatures between 21.6°C and $38.4^\circ\text{C} = 49.85\% + 34\% = 83.85\%$.

✓ (3 marks)

22 i 1

(1 mark)

ii 1



If 2.5% of girls are taller than Rachel, the corresponding z-score is 2.

From the graph, this is a height of 155 cm. Hence, Rachel is 155 cm tall.

(1 mark)

- 2 At age $15\frac{1}{2}$, 155 cm is a z-score of -1.

$\therefore$ From a z-score of 0 to -1, the percentage of girls is 34%.

$\therefore (34 + 50)\% = 84\%$ of girls are taller than Rachel.

✓ (2 marks)

- iii Average height is a z-score of 0.

$\therefore$ Average height is 163 cm.

(1 mark)

iv $B = \frac{m}{h^2}$

$$B = 21 \text{ and } h = 1.63$$

$$\therefore 21 = \frac{m}{(1.63)^2}$$

$$\therefore m = 21 \times (1.63)^2$$

$$= 55.7949$$

$$\approx 56 \text{ kg}$$

✓ (2 marks)

23 i $z = \frac{x - \mu}{\sigma}$

$$\text{Test 1: } z = \frac{63 - 60}{6.2}$$

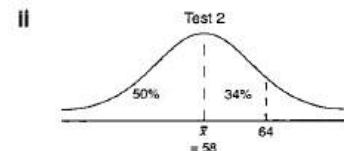
$$= 0.4838 \dots$$

$$\text{Test 2: } z = \frac{62 - 58}{6}$$

$$= 0.6666 \dots$$

Stuart performed better in Test 2 as his z-score is higher.

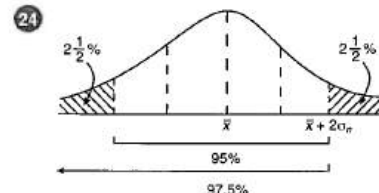
✓ (2 marks)



You would expect 84% of students to score less than 64.

$$\text{i.e. } \frac{84}{100} \times 150 = 126 \text{ students.}$$

✓ (2 marks)



If 97.5% last 5000 hours,

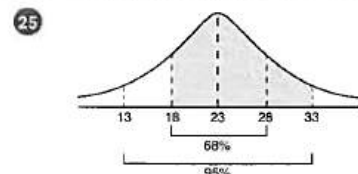
$$\text{then } 5000 = \mu + 2\sigma$$

$$5000 = \mu + 2 \times 170$$

$$\mu = 5000 - 2 \times 170$$

$$= 4660 \text{ hours}$$

(1 mark)



Now $(95\% - 68\%) \div 2 = 13.5\%$

$\therefore$ Percentage of scores between 18 and 33

$$= 95\% - 13.5\%$$

$$= 81.5\%$$

Answer D

26 i $\mu = 754 \text{ g}, \sigma = 2 \text{ g}$

$$z = \frac{x - \mu}{\sigma} = \frac{754 - 754}{2} = 0$$

(This can be observed from the graph drawn in (iii)).

(1 mark)

ii $-1 = \frac{x - 754}{2}$

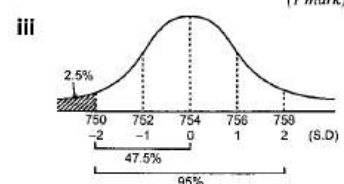
$$-2 = x - 754$$

$$x = -2 + 754$$

$$= 752$$

$\therefore$ Weight of box is 752 g.

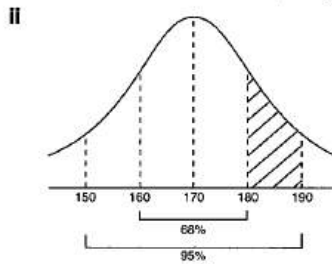
(1 mark)



95% of the weights are within 2 standard deviations.

$\therefore (95 \div 2)\% = 47.5\%$ are between 750 g and 754 g. ✓
 $\therefore 50\% + 47.5\% = 97.5\%$ are greater than 750 g.
 $\therefore 2.5\%$ of boxes will have a weight less than 750 g. ✓
 (2 marks)

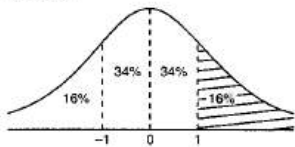
- 27 i Roberto's z-score could be any number between 1 and 2 as this shaded region is the 2nd standard deviation from the mean. (1 mark)



Percentage of data in shaded region
 $\div (95 - 68) \div 2$ ✓
 $\div 13.5\%$. ✓
 (2 marks)

- 28 i $z = \frac{x - \mu}{\sigma}$
 A z-score of 1 means that Hardev's score is exactly one standard deviation from the mean. For example, if Hardev's score was 80 and the mean was 60 and the standard deviation was 20 then,
 $z = \frac{80 - 60}{20} = 1$ ✓
 (2 marks)

- ii About 68% of scores lie within a z-score range of -1 and 1.
 $\therefore$ About 34% lie within a z-score range of 0 and 1.
 $\therefore$ Proportion of people with higher score than Hardev
 $\div 100\% - 50\% - 34\%$
 $\div 16\%$

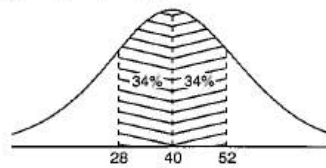


(1 mark)

- 29 $z = \frac{x - \mu}{\sigma}$
 $3 = \frac{x - 55}{6}$
 $18 = x - 55$
 $18 + 55 = x$
 $\therefore x = 73$

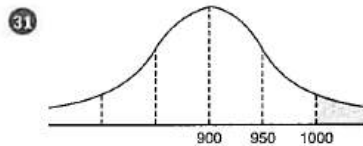
Answer D

- 30 $\mu = 40$ $\sigma = 12$



50% are younger than 40.
 Appox. 68% lie within one standard deviation, i.e. between 28 and 52.
 $\therefore 34\%$ are between 40 and 52.
 $\therefore$ Percentage younger than 52
 $= 50\% + 34\%$
 $= 84\%$

Answer D



Shaded region $= (100\% - 95\%) \div 2$
 $= 2.5\%$

Answer B

- 32 i z-score = -2
 Kim's score was 2 standard deviations below the mean. ✓✓
 (2 marks)

- ii Actual mark
 $= 75 - 2 \times 5$
 $= 65$
 or, using the formula:

$$z = \frac{x - \mu}{\sigma}$$

$$-2 = \frac{x - 75}{5}$$

$$-10 = x - 75$$

$$-10 + 75 = x$$

$$x = 65$$

(1 mark)